

2018

COMPUTER SCIENCE – GENERAL

Second Paper

Full Marks : 100

Group - A

(Algorithms and Data Structure)

2. (a) Write down the algorithm of binary search and mention its worst case complexity. 5+2
(b) State the advantages of linked allocation over contiguous allocation. 4
(c) State the characteristics of an algorithm. What do you mean by complexity of an algorithm? 3+2
3. (a) What is singly linked list? 2
(b) Write an algorithm for insertion sort to arrange the element of an array in ascending order. 4
(c) What is a flow chart? Draw the symbols used in a flow chart stating their purposes. Draw a flow chart to find sum of all odd numbers upto N, where N is input from user. 2+2+4
(d) Explain array with an example. 2
4. (a) State differences between connected graphs and unconnected graphs with examples and diagrams. 5
(b) Write separate algorithms for the following queue operations :
(i) Test whether queue is FULL.
(ii) Test whether queue is empty.
(iii) Insert an element to the rear of the queue.
(iv) Remove an element from the front of the queue. 1+1+2+2
(c) Construct Binary Search Tree using the following nodes : 5
M P T F O Z R J L A